Type MS · What algorithms are and how we write them down

Answer 1 of 6 · Question 6 part a · 2 marks

2023 JUN · Paper 01 · extract 1 of 5 · find part a in this mark scheme extract

6(a)	Award **one** mark for each of: Method • Flowchart (1) Justification • It is a visual representation / does not use many words / does not rely on use of English language (1) • It does not rely on understanding specific syntax (1) • Overview without unnecessary detail (1) OR Method • Pseudocode/structured English/written description (1) Justification • It does not rely on understanding specific syntax (1) • It uses English language words (1)	Valid method must be specified first before mark for justification can be awarded.	2

Type MS · What algorithms are and how we write them down

Answer 1 of 6 · Question 6 part a · 2 marks

2023 JUN · Paper 01 · extract 2 of 5 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)	Award **one** mark for any of the following: • The test on line 3 has already limited the values to 1 and 2 (1) • The only value left for inNum to be (on line 7) is 2 (1)		1

Question Number	Answer	Additional Guidance	Mark
6(c)(i)	Award up to **two** marks for a linked description, such as: • The oldIndex value will go from highest to lowest (1) the step value is negative/decremented by 1 (each iteration) (1)	Allow 4 to 0 instead of highest and lowest	2

Question Number	Answer	Additional Guidance	Mark
6(c)(ii)	Award **one** mark for any of the following: • Reverse the (array of) scores (1) • Copy the oldScores into newScores, in reverse order (1)		1

Type MS · What algorithms are and how we write them down

Answer 1 of 6 · Question 6 part a · 2 marks

2023 JUN · Paper 01 · extract 3 of 5 · find part a in this mark scheme extract

Question Number	Indicative Content	Additional Guidance	Mark
6(d)	**Growth** • Higher bandwidth available both on wired and wireless networks • Reduced costs of large capacity storage devices • Improved levels of encryption / authentication • Increased trust in service providers • Growth of service provider options • Increased business models that aid scalability regarding storage capacity and costs **Benefits** • Sharing / collaboration on documents and setting of access permissions • Access from any location connected to a WAN • Reduced cost of local computing devices with less storage required • Built in backups of data on server-based systems **Drawbacks** • Increased security risks incurred from outsourcing a service • Relies on high-speed network connections • Reliance on service providers continuing to provide a high level of service • Increased chances of external data breaches • Identifying responsibilities for data breaches / legal jurisdiction • Increased power consumption from server systems and associated cooling systems • Increased use of networks and network traffic which will have an impact on other network users **Conclusion** • A conclusion does not have to be stated explicitly, but can be inferred from the discussion		6

Type MS · What algorithms are and how we write them down

Answer 1 of 6 · Question 6 part a · 2 marks

2023 JUN · Paper 01 · extract 4 of 5 · find part a in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

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Answer 1 of 6 · Question 6 part a · 2 marks

2023 JUN · Paper 01 · extract 5 of 5 · find part a in this mark scheme extract

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Type MS · What algorithms are and how we write them down

Answer 2 of 6 · Question 4 part a · 1 mark

2021 NOV · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

<table>
<tr>
<td>4(a)</td>
<td>Award one mark from:

• a step-by-step description of a process that completes a task (1)
• a set of instructions that describes how to get something done (1)</td>
<td></td>
<td>1</td>
</tr>
<tr>
<td>4(b)(i)</td>
<td>Award one mark each to a maximum of five marks for:

• two correct decisions in last two lines (1)
• two correct outputs (1)
• correct increment at Count=Count+1 (1)
• correct change for Number3=Number2+Number1 (1)
• correct change for Number1=Number2 AND Number2=Number3 (1)

<table>
<tr><th>Number 1</th><th>Number 2</th><th>Number 3</th><th>Count</th><th>Output</th><th>Count = 2?</th></tr>
<tr><td>1</td><td>1</td><td>-</td><td>0</td><td>1 1</td><td>False</td></tr>
<tr><td>1</td><td>2</td><td>2</td><td>1</td><td>2</td><td>False</td></tr>
<tr><td>2</td><td>3</td><td>3</td><td>2</td><td>3</td><td>True</td></tr>
</table>
</td>
<td>Ignore spelling errors

Allow true false, Y N etc. for decisions</td>
<td>5</td>
</tr>
</table>

Type MS · What algorithms are and how we write them down

Answer 2 of 6 · Question 4 part a · 1 mark

2021 NOV · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

4(b)(ii)	Award **one** mark each for: • helps visualise how the algorithm works (1) • helps detect (logic) errors (1)		1

Question Number	Answer	Additional Guidance	Mark
4(c)	Award **one** mark each to a maximum of **five** marks for: • SET count TO 0 (1) • IF statement checks for a match to 1 vowel (1) • IF statement checks for a match to all vowels (1) • count incremented correctly (1) • display count (1) • concatenates suitable message with the count (1) SET word TO "elephant" SET count TO 0 FOR EACH letter FROM word DO IF letter = 'a' OR letter = 'e' OR letter = 'i' OR letter = 'o' OR letter = 'u' THEN SET count TO count + 1 END IF END FOREACH SEND 'The number of vowels is ' & count TO DISPLAY		5
		Total	**12**

Type MS · What algorithms are and how we write them down

Answer 2 of 6 · Question 4 part a · 1 mark

2021 NOV · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · What algorithms are and how we write them down

Answer 3 of 6 · Question 4 part b.ii · 1 mark

2021 NOV · Paper 01 · extract 1 of 3 · find part b.ii in this mark scheme extract

<table>
<tr>
<td>4(a)</td>
<td>Award one mark from:

• a step-by-step description of a process that completes a task (1)
• a set of instructions that describes how to get something done (1)</td>
<td></td>
<td>1</td>
</tr>
<tr>
<td>4(b)(i)</td>
<td>Award one mark each to a maximum of five marks for:

• two correct decisions in last two lines (1)
• two correct outputs (1)
• correct increment at Count=Count+1 (1)
• correct change for Number3=Number2+Number1 (1)
• correct change for Number1=Number2 AND Number2=Number3 (1)

<table>
<tr><th>Number 1</th><th>Number 2</th><th>Number 3</th><th>Count</th><th>Output</th><th>Count = 2?</th></tr>
<tr><td>1</td><td>1</td><td>-</td><td>0</td><td>1 1</td><td>False</td></tr>
<tr><td>1</td><td>2</td><td>2</td><td>1</td><td>2</td><td>False</td></tr>
<tr><td>2</td><td>3</td><td>3</td><td>2</td><td>3</td><td>True</td></tr>
</table>
</td>
<td>Ignore spelling errors

Allow true false, Y N etc. for decisions</td>
<td>5</td>
</tr>
</table>

Type MS · What algorithms are and how we write them down

Answer 3 of 6 · Question 4 part b.ii · 1 mark

2021 NOV · Paper 01 · extract 2 of 3 · find part b.ii in this mark scheme extract

4(b)(ii)	Award **one** mark each for: • helps visualise how the algorithm works (1) • helps detect (logic) errors (1)		1

Question Number	Answer	Additional Guidance	Mark
4(c)	Award **one** mark each to a maximum of **five** marks for: • SET count TO 0 (1) • IF statement checks for a match to 1 vowel (1) • IF statement checks for a match to all vowels (1) • count incremented correctly (1) • display count (1) • concatenates suitable message with the count (1) SET word TO "elephant" SET count TO 0 FOR EACH letter FROM word DO IF letter = 'a' OR letter = 'e' OR letter = 'i' OR letter = 'o' OR letter = 'u' THEN SET count TO count + 1 END IF END FOREACH SEND 'The number of vowels is ' & count TO DISPLAY		5
		Total	**12**

Type MS · What algorithms are and how we write them down

Answer 3 of 6 · Question 4 part b.ii · 1 mark

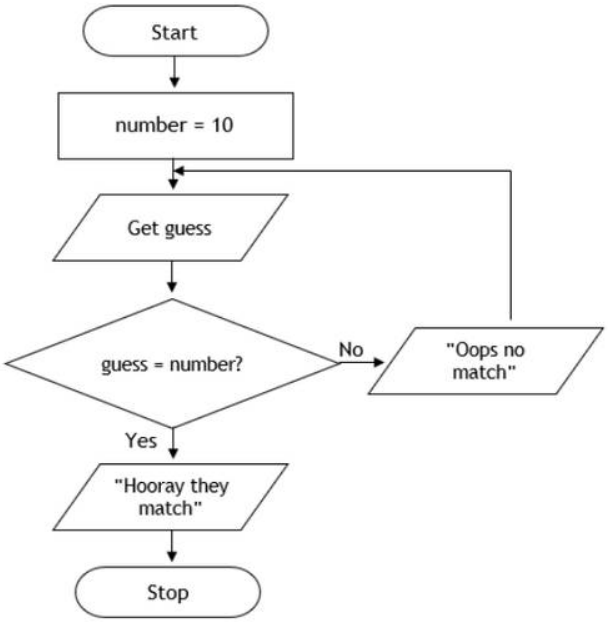
2021 NOV · Paper 01 · extract 3 of 3 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · What algorithms are and how we write them down

Answer 4 of 6 · Question 4 part a.ii · 1 mark

2022 JUN · Paper 01 · extract 1 of 3 · find part a.ii in this mark scheme extract

4(a)(i)	Award up to **five** marks for: • Start and stop terminators in the correct positions (1) 1-3 • Number set to 10 AND Get guess in the correct positions (1) 4-6 • Loop back to before Get guess and after number set to 10 if there is no match (1)Does not need to go via message box • Yes/no labels on decision match output messages (1) 4-6 • Correctly connected as in MS image, with at least 6 arrows correct (1) 7-9  	Boxes should be marked by content rather than shape.	5
4(a)(ii)	**The only correct answer is C** *A is not correct because a simulation is a completed program* *B is not correct because a cipher is a form of encryption* *D is not correct because a truth table is a method of testing an algorithm*		1

Type MS · What algorithms are and how we write them down

Answer 4 of 6 · Question 4 part a.ii · 1 mark

2022 JUN · Paper 01 · extract 2 of 3 · find part a.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(b)(i)	Award **one** mark for: • D		1
4(b)(ii)	Award **one** mark for: • B		1

Question Number	Answer	Additional Guidance	Mark
4(c)(i)	Award up to **four** marks for: • RedPoints column correct (1) • OrangePoints AND NumOranges columns correct (1) • Score correct in row 8 OR row 9 (1) -1 or below • Outputs correct and starting on the same row as the score OR the row below (1) can all be on one line		4
4(c)(ii)	Award **one** mark for: • 23		1

Colour	Score	Red Points	Orange Points	Num Oranges	Outputs
	0	0	0	0	
red		1			
orange			8	1	
red		2			
red		3			
orange			16	2	
-1					
	19				
					Score: 19
					Number of reds: 3
					Number of oranges: 16

Type MS · What algorithms are and how we write them down

Answer 4 of 6 · Question 4 part a.ii · 1 mark

2022 JUN · Paper 01 · extract 3 of 3 · find part a.ii in this mark scheme extract

4(c)(iii)	Award **one** mark for: • Pseudododode that replaces OrangePoints with NumOranges on line 23 (1) SEND ("Number of oranges: "& NumOranges) TO DISPLAY		1
		Total for question 4	**14**

Question Number	Answer	Additional Guidance	Mark

Type MS · What algorithms are and how we write them down

Answer 5 of 6 · Question 3 part a · 2 marks

2023 JUN · Paper 02 · extract 1 of 6 · find part a in this mark scheme extract

3(a)	A1 A2	Award **one** mark for any of the following up to a maximum of **two** marks: • Each column represents a variable/output (1) • Rows shows how the values of variables change (as the code is executed) (1) • Values are filled in manually/by hand (1)	Line numbers are for guidance – mark answer as a whole	**2**

Question	mp	Answer	Additional Guidance	Mark
3(b)	B1 B2 B3	Award **one** mark for each correct cell: **num1** | **num2** | **Output** 88 | 18 | State 2 17 | 18 | State 4 12 | 19 | State 1	ignore quotes e.g. "State 2"	**3**

Type MS · What algorithms are and how we write them down

Answer 5 of 6 · Question 3 part a · 2 marks

2023 JUN · Paper 02 · extract 2 of 6 · find part a in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark
3(c)	Award **one** mark for each of:		C4 should exit loop Allow C5 if a loop is used to keep asking for a non-negative exponent C9 – ignore error message in C6; must have base, exponent, and answer in output for this mark	**10**
	C1	At least one variable with a suitable variable name		
	C2	At least one variable initialised to an appropriate value		
	C3	While loop used to continue running code		
	C4	Relational test to identify terminating condition for loop		
	C5	Selection used to check for negative exponent		
	C6	Appropriate error message if exponent is negative		
	C7	Calculation of answer using any method		
	C8	A loop used to calculate answer as indicated in flowchart		
	C9	Prompts and output messages are fit for purpose		
	C10	Executing and producing the correct output for anticipated inputs		

Suggested test data and output		
Base	**Exponent**	**Output**
0		ends program
2	0	2 to power of 0 is 1
3	4	3 to power of 4 is 81

Type MS · What algorithms are and how we write them down

Answer 5 of 6 · Question 3 part a · 2 marks

2023 JUN · Paper 02 · extract 3 of 6 · find part a in this mark scheme extract

C#

```
// Initialise variables
int baseNum = 0;
int exponent = 0;
int answer = 1;
String outString = "";

// Display the first prompt and get the base number
Console.Write("Enter a base number (0 to exit)): ");
baseNum = Convert.ToInt32(Console.ReadLine());

// Check if user wants to exit
while (baseNum != 0.0)
{
    // Display a prompt and get the exponent number
    Console.Write("Enter an exponent number: ");
    exponent = Convert.ToInt32(Console.ReadLine());

    // Check if the exponent is negative
    if (exponent < 0)
    {
        Console.Write("Invalid exponent entered");
    }
    else
    {
        answer = 1;

        // Calculate the answer using exponentiation
        for (int i = 0; i < exponent; i++)
        {
            answer = answer * baseNum;
        }

        // Create a string for the output and show the user
        outString = baseNum.ToString() + " to the power of " +
                    exponent.ToString() + " is " +
                    answer.ToString();
        Console.Write(outString);
    }
    // Display the first prompt again and get the base number
    Console.Write("\nEnter a base number (0 to exit)): ");
    baseNum = Convert.ToInt32(Console.ReadLine());
}
```

Type MS · What algorithms are and how we write them down

Answer 5 of 6 · Question 3 part a · 2 marks

2023 JUN · Paper 02 · extract 4 of 6 · find part a in this mark scheme extract

Java

```
// --------------------------------------------------------
// Write your code below this line

// Initialise variables
int baseNum = 0;
int exponent = 0;
int answer = 1;
String outString = "";

Scanner myScanner = new Scanner (System.in);

// Display the first prompt and get the base number
System.out.print("Enter a base number (0 to exit)): ");
baseNum = Integer.parseInt (myScanner.nextLine());

// Check if user wants to exit
while (baseNum != 0.0)
{
    // Display a prompt and get the exponent number
    System.out.print("Enter an exponent number: ");
    exponent = Integer.parseInt (myScanner.nextLine());

    // Check if the exponent is negative
    if (exponent < 0)
    {
        System.out.println ("Invalid exponent entered");
    }
    else
    {
        answer = 1;

        // Calculate the answer using exponentiation
        for (int i=0; i<exponent; i++)
        {
            answer = answer * baseNum;
        }

        // Create a string for the output and show the user
        outString = Integer.toString(baseNum) + " to the power of " +
                    Integer.toString(exponent) + " is " +
                    Integer.toString(answer);
        System.out.println(outString);
    }

    // Display the first prompt again and get the base number
    System.out.print("Enter a base number (0 to exit)): ");
    baseNum = Integer.parseInt(myScanner.nextLine());
}
```

Type MS · What algorithms are and how we write them down

Answer 5 of 6 · Question 3 part a · 2 marks

2023 JUN · Paper 02 · extract 5 of 6 · find part a in this mark scheme extract

Python

```
# Initialise variables
base = 0
exponent = 0
answer = 1

# Get the first base number
base = int (input ("Enter a base number (0 to exit): "))

# Check if the user wants to exit
while (base != 0):

    # Get the exponent number
    exponent = int (input ("Enter an exponent number: "))

    # Check if the exponent is negative
    if (exponent < 0):
        print ("Invalid exponent entered")
    else:
        # Reset the answer
        answer = 1

        # Calculate the answer using exponentiation
        for count in range (0, exponent):
            answer = answer * base

        # Create a string for the output sentence
        print (str (base) + " to the power of " +
               str (exponent) + " is " +
               str (answer))

    # Get another base number and lp
    base = int (input ("Enter a base number (0 to exit): "))
```

Type MS · What algorithms are and how we write them down

Answer 5 of 6 · Question 3 part a · 2 marks

2023 JUN · Paper 02 · extract 6 of 6 · find part a in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · What algorithms are and how we write them down

Answer 6 of 6 · Question 2 part a.i · 1 mark

2024 JUN · Paper 02 · extract 1 of 14 · find part a.i in this mark scheme extract

2(a)(i)	A1	**The only correct answer is A** ***B*** *is not correct because* ***computational thinking*** *are skills used in creating algorithms and programs* ***C*** *is not correct because* ***decomposition*** *is breaking down a problem into smaller parts* ***D*** *is not correct because* ***pseudocode*** *is a notation for expressing an algorithm*		**1**

Type MS · What algorithms are and how we write them down

Answer 6 of 6 · Question 2 part a.i · 1 mark

2024 JUN · Paper 02 · extract 2 of 14 · find part a.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · What algorithms are and how we write them down

Answer 6 of 6 · Question 2 part a.i · 1 mark

2024 JUN · Paper 02 · extract 3 of 14 · find part a.i in this mark scheme extract

2(a)(ii)	A2	Award **one** mark for: • Variable (1)	Allow parameter Ignore local/global	**1**

Question	mp	Answer	Additional Guidance	Mark

Type MS · What algorithms are and how we write them down

Answer 6 of 6 · Question 2 part a.i · 1 mark

2024 JUN · Paper 02 · extract 4 of 14 · find part a.i in this mark scheme extract

2(b)(i)	B1	Award **one** mark for any of the following: • An instruction is impossible to execute (1) • The program stops executing **unexpectedly** / **crashes** (1)	Do not award an example, as that is the next question Don't award repeat of question e.g. "Stopping during execution" without clarifying reason or that stop was unexpected	**1**

Question	mp	Answer	Additional Guidance	Mark

Type MS · What algorithms are and how we write them down

Answer 6 of 6 · Question 2 part a.i · 1 mark

2024 JUN · Paper 02 · extract 5 of 14 · find part a.i in this mark scheme extract

2(b)(ii)	B2	Award **one** mark for any of the following: • Division by zero (1) • Accessing memory outside the range of an array (1) • Attempting an operation on incompatible types (1) • File not found (1) • An exception is raised that cannot be/is not handled (1)	Award other examples, if they can be mapped to a bullet	**1**

Type MS · What algorithms are and how we write them down

Answer 6 of 6 · Question 2 part a.i · 1 mark

2024 JUN · Paper 02 · extract 6 of 14 · find part a.i in this mark scheme extract

Type MS · What algorithms are and how we write them down

Answer 6 of 6 · Question 2 part a.i · 1 mark

2024 JUN · Paper 02 · extract 7 of 14 · find part a.i in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · What algorithms are and how we write them down

Answer 6 of 6 · Question 2 part a.i · 1 mark

2024 JUN · Paper 02 · extract 8 of 14 · find part a.i in this mark scheme extract

2(c)		Award **one** mark for each of:		
	C1	Add or remove a semicolon on initialisation of the length variable (1)	• `int length = 4;` • `int length = 4;` • `length = 4`	
	C2	Add a comma between the arguments to the subprogram call to generate a random number (1)	• `width = randGenerator.Next (1, 6);` • `width = randGenerator.nextInt(1, 6);` • `width = random.randint (1, 5)`	
	C3	Replace modulus operator (%) with multiplication operator (*) (1)	• `perimeter = (2 * width) + (2 * length);` • `perimeter = (2 * width) + (2 * length);` • `perimeter = (2 * width) + (2 * length)`	**3**

Type MS · What algorithms are and how we write them down

Answer 6 of 6 · Question 2 part a.i · 1 mark

2024 JUN · Paper 02 · extract 9 of 14 · find part a.i in this mark scheme extract

C#

```
// Q02cFINISHED

int width = 0;
int length = 4;
int perimeter = 0;
Random randGenerator = new Random();

width = randGenerator.Next (1, 6);
perimeter = (2 * width) + (2 * length);

Console.WriteLine (width.ToString() + " " +
                   length.ToString() + " " +
                   perimeter.ToString());
```

Type MS · What algorithms are and how we write them down

Answer 6 of 6 · Question 2 part a.i · 1 mark

2024 JUN · Paper 02 · extract 10 of 14 · find part a.i in this mark scheme extract

Java

```java
// Q02cFINISHED

import java.util.Random;

public class Q02cFINISHED
{
    public static void main (String[] args)
    {
        int width = 0;
        int length = 4;
        int perimeter = 0;
        Random randGenerator = new Random();

        width = randGenerator.nextInt(1, 6);
        perimeter = (2 * width) + (2 * length);

        System.out.println (String.valueOf(width) + " " +
                            String.valueOf(length) + " " +
                            String.valueOf(perimeter));
    }
}
```

Type MS · What algorithms are and how we write them down

Answer 6 of 6 · Question 2 part a.i · 1 mark

2024 JUN · Paper 02 · extract 11 of 14 · find part a.i in this mark scheme extract

Python

```
# Q02cFINISHED

import random

width = 0
length = 4
perimeter = 0

width = random.randint (1, 5)
perimeter = (2 * width) + (2 * length)

print (width, length, perimeter)

```

Type MS · What algorithms are and how we write them down

Answer 6 of 6 · Question 2 part a.i · 1 mark

2024 JUN · Paper 02 · extract 12 of 14 · find part a.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · What algorithms are and how we write them down

Answer 6 of 6 · Question 2 part a.i · 1 mark

2024 JUN · Paper 02 · extract 13 of 14 · find part a.i in this mark scheme extract

<table>
<tr>
<td>2(d)</td>
<td>D1
D2
D3
D4</td>
<td>
<table>
<tr><td>Erroneous</td><td>Award one mark for any of the following up to a maximum of two marks:
Contains incorrect data type (1)
<2 letters (1)
9+ letters (1)</td></tr>
<tr><td>Normal</td><td>A string between 2 and 8 in length that includes only letters (1)</td></tr>
<tr><td>Boundary</td><td>Award one mark for any of the following:
A string of exactly 8 letters (1)
A string of exactly 2 letters (1)</td></tr>
</table>
</td>
<td>Award any correct example (not description)
Do not award an empty cell as equivalent to empty string
Do not accept same test twice for erroneous</td>
<td>4</td>
</tr>
</table>

Type MS · What algorithms are and how we write them down

Answer 6 of 6 · Question 2 part a.i · 1 mark

2024 JUN · Paper 02 · extract 14 of 14 · find part a.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark